



United International University

Department of Computer Science and Engineering

CSE 3313: Computer Architecture

Midterm Examination: Fall 2025

Total Marks: 30 Time: 1 hour and 30 minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all questions. Numbers to the right of the questions denote their marks.

1. A program P runs on computer A with a clock frequency of 2 GHz. The instructions of the program are classified with the following distribution:

Instruction Type	R Type	I Type	J Type
Instruction Count	10	5	4
Cycles Per Instruction	3	6	2

Table 1: Program P Description

- (a) If program P completes its entire execution after 50 ns, find out the affected and unaffected time. [4]
(b) A second computer B, with the same Instruction Set Architecture as that of A, wants to run program P in half of the execution time as that of A. Calculate the required clock frequency of computer B. [3]
(c) Find out the improvement factor if the execution of R type instructions can be improved by 2 times. [3]
2. (a) Consider the following C function. The arguments will be found at \$a0-\$a2. Convert the code to the corresponding MIPS assembly instructions. [8]

```
1  int my_function(int a, int b, int arr [ ] )
2  {
3      int sum;
4      while(b >= a)
5      {
6          sum += ~ ( arr [ b ] * 16 ) % 32 );
7          a++;
8      }
9      return sum;
10 }
```

Code for 2 (a)

- (b) Observe the following MIPS code and convert it into machine code. Conversion of the machine code to binary is not required. [3]

```
1  100: my_function:
2  104: bne $s0, $t8, IF
3  108: sub $t3, $s1, $s5
4  112: ori $t2, $t7, 42
5  116: lw $t1, 20 ( $t2)
6  120: IF:
7  124: slt $t1, $s2, $s5
8  128: j my_function
```

Code for 2 (b)

- (c) Assume that processor P has 256 registers. The size of MIPS instruction is 32 bits and 6 bits are reserved for opcode. A computer architect wrote the following C instruction that he wants to process in processor P. Justify the errors in the code the architect may face while converting to MIPS and then binary. [5]

```
1  int main()
2  {
3      int arr [ 301 ] = {0};
4      int c = arr [ 0 ] & 1024;
5  }
```

Code for 2 (c)

3. Use optimized division algorithm to divide two numbers A by B, where A = 13 and B = 6.

[4]

Instruction	Opcode	Function Code
add	0	32
sub	0	34
lw	35	-
sw	43	-
and	0	36
or	0	37
nor	0	39
andi	12	-
ori	13	-
sll	0	0
srl	0	2
beq	4	-
bne	5	-
slt	0	42
j	2	-
jr	0	8
jal	3	-
addi	8	-

Table 1: MIPS Machine Codes

Name	Register Number
\$zero	0
\$at	1
\$v0-\$v1	2-3
\$a0-\$a3	4-7
\$t0-\$t7	8-15
\$s0-\$s7	16-23
\$t8-\$t9	24-25
\$k0-\$k1	26-27
\$gp	28
\$sp	29
\$fp	30
\$ra	31

Table 2: MIPS Registers